



Links

LinkedIn:

[linkedin.com/in/rajarajeswari-mani](https://www.linkedin.com/in/rajarajeswari-mani)

Skills

Programming:

C, C++, Python, CAPL

Automotive Cybersecurity:

TARA, ATA, UNECE R155/156, STRIDE, Secure Boot, SecureAccess, SecureJTAG, HSM, OTA/SOTA

Cryptography:

AES, RSA, ECC, ECDH/ECDSA, PKI/KMS

Functional Safety (ISO 26262):

HARA, ASIL, FMEA, FTA, DFA, FMEDA

AUTOSAR Classic:

Crypto Stack (CSM/CRY/Crylf/SecOC), DCM, UDS 14229-1

Automotive Tools:

Vector DaVinci, CANoe/CANalyzer, TRACE32, SIL/HIL

Protocols:

CAN, CAN FD, UDS, Ethernet, DoIP, SPI, I2C

Standards & Process:

ASPICE, V-Model, Agile/Scrum, MISRA C:2012

MBSE & Requirements:

SysML/UML, DOORS, Rhapsody, Polaron

Vulnerability & Risk:

VARA, CVE/CWE, CVSS, SCA, SBOM, SAST

Rajarajeswari Mani

AUTOMOTIVE CYBERSECURITY & FUNCTIONAL SAFETY ARCHITECT

Contact

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Personal Information

- **Nationality:** Indian
- **Education:** B.E. Electronics and Communication Engineering — Chennai, 2010
- **Languages:** English (C1), German (A1), Tamil (Native, C2)

About Me

- A career shaped through steady growth - from hands-on firmware development, through embedded application engineering, into project and technical management, and now technical/architectural leadership.
- 14+ years in automotive embedded software, including 7+ years in automotive cybersecurity (ISO/SAE 21434) and functional safety (ISO 26262) across ADAS/AD, AUTOSAR, and BMS.
- TÜV SÜD-certified in ISO/SAE 21434 and ISO 26262; ISTQB-certified (CTFL); deep expertise in TARA/VARA, secure embedded architecture, and cryptography.
- Bridges cybersecurity and functional safety domains, reconciling ASIL/CAL classification and HARA/TARA outcomes across cross-functional teams.

Core Competencies

- **BMW** **BMW (ADAS/AD, AUTOSAR):** TARA per ISO/SAE 21434, Secure Boot, SecureJTAG, SecureAccess, AUTOSAR Crypto Stack (CSM/Crylf/SecOC), HSM, UDS Security Access; ASIL/CAL alignment, CVE/CVSS tracking.
- **VW / AUDI** **Volkswagen/Audi (Variant Coding & Diagnostics):** UDS seed-key interfaces, MISRA-C/CERT-C; SFD, KGAS, CSGA gate reviews as Scrum Master.
- **HYUNDAI KIA** **Hyundai Kia Motors (Camera ECU):** AUTOSAR BSW/DCM, Lauterbach TRACE32, CAPL/CANoe/CANalyzer; on-site integration in Seoul (ImageNext).
- **EDAG** **EDAG (BMS, Tier-1 Supplier):** ISO 26262 HARA/ASIL, FMEA/FTA/DFA, FMEDA metrics; safety evidence for certification.
- **CROSS-OEM** **Cross-OEM:** ASPICE, V-Model, Agile, MBSE (SysML/UML, DOORS, Rhapsody), traceability, SEOOc compliance.

Certifications

- TÜV SÜD Certified Automotive Cyber Security Practitioner (CACSP)
- TÜV SÜD Certified Level 1 ISO 26262 Functional Safety Practitioner (FSCP)
- ISTQB Certified Tester Foundation Level (CTFL)

Current Role Focus

- Automotive Cybersecurity Architect - ISO/SAE 21434 TARA & Secure Architecture
- Functional Safety Engineer - ISO 26262 HARA/ASIL, FMEA/FTA
- AUTOSAR Security Engineer - Crypto Stack, HSM, SecureAccess, SecureJTAG, Secure Diagnostics

Work Experience

Lead Software Engineer - BMS & AUTOSAR Crypto Stack (BMW-compliant ADAS/AD)

EDAG Engineering GmbH - Munich, Germany

Jun 2023 - Jan 2026

- Led safety-oriented design for lithium-ion BMS functions per ISO 26262 - HARA/ASIL determination, FMEA/FTA/DFA analysis, and CANTATA-based unit testing - and compiled safety evidence for certification submission.
- Led system-level TARA per ISO/SAE 21434 for a BMW-compliant AUTOSAR ADAS ECU, translating OEM goals into system-specific mitigations via Security Architecture Reviews aligned with CSMS.
- Architected SecureAccess, SecureJTAG/Debug, Secure/Authentic Boot, RTMD, and secure flashing controls; configured the AUTOSAR Crypto Stack (Vector DaVinci) with KeyM and CAL-rated SecOC for ASIL D/C signals plus certificate-based UDS Authentication (0x29).
- Enabled HSM-based key storage, secure logging, and OTA/FOTA/SOTA readiness; validated STRIDE-classified threat resilience via fuzz/penetration testing.
- Coordinated with Functional Safety teams to reconcile ISO 26262/ISO 21434 outcomes across BMS and ADAS/AD programs.

Associate Project Manager - ADAS Integration & Variant Coding/Diagnostics

Bosch Global Software Technologies - Coimbatore, India

Feb 2016 - Jan 2023

- Owned component-level TARA for BMW AUTOSAR ADAS ECUs, maintaining bidirectional traceability in IBM DOORS per ISO/SAE 21434 and UNECE R155/R156.
- Configured AUTOSAR Crypto Stack (CSM/Crylf) for SecOC/AES-CMAC secure CAN communication, protecting ASIL D/C signals against replay attacks; integrated HSM services and SecureAccess (UDS 0x27)/SecureJTAG-Debug controls.
- Aligned ISO 26262/ISO 21434 via joint ASIL/CAL reviews; performed HARA, DFA, and FMEDA-based hardware safety metrics with fault injection testing; tracked CVE/CWE and CVSS findings via CANoe.
- Led as Sub-Project Manager/Scrum Master driving agile delivery and mentorship; developed UDS seed-key diagnostic interfaces (MISRA-C/CERT-C) for Volkswagen/Audi variant-coding programs, supporting SFD, KGAS, and CSGA security-gate reviews.

Software Engineer - AUTOSAR BSW Engineering (Camera ECU, AISIN and ImageNext)

KPIT Technologies Ltd - Bangalore, India

Sep 2013 - Jan 2016

- Configured AUTOSAR BSW stack modules and validated DCM diagnostics via Lauterbach TRACE32, Vector UDE, and CAPL scripts in CANoe/CANalyzer.
- Resolved build/integration issues, ensuring MISRA C:2012-compliant, quality-assured software deliverables.
- Completed a 3-month on-site deployment to Seoul (ImageNext) for AUTOSAR BSW integration on Hyundai Kia camera ECUs.

Software Developer - Real-time Industrial Projects

VI Microsystems Pvt. Ltd - Chennai, India

Mar 2011 - Jul 2013

- Built firmware on simulated platforms and applied FPGA and VHDL-based testing strategies to maintain performance and reliability in real-time industrial embedded systems.